

Name: _____ Date: _____

Period: _____

Spiral Review — Prerequisites for Lesson 3.10

1. Evaluate each inverse trigonometric expression. Give your answer in radians.

(a) $\arcsin\left(\frac{\sqrt{2}}{2}\right) = \frac{\pi}{4}$ (b) $\arccos\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$

2. Given that $\sin \theta = \frac{1}{2}$, list **all** values of θ in the interval $[0, 2\pi)$ that satisfy this equation.

$$\theta = \frac{\pi}{6} \text{ and } \frac{5\pi}{6}$$

Explain briefly why there are two solutions (not just one): **Sine is positive in both Quadrant I and Quadrant II. The reference angle $\frac{\pi}{6}$ gives one solution in Q1; the supplementary angle $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$ gives the Q2 solution.**

3. Solve for $\sin x$ by isolating the trig function. **Do not** find the actual angle — just isolate $\sin x$.

$$2\sin x + 1 = 0$$

$$\sin x = -\frac{1}{2}$$

4. Which of the following expressions gives the value of θ in Quadrant II if $\sin \theta = \frac{3}{5}$? Circle the correct answer.

$$\theta = \arcsin\left(\frac{3}{5}\right) \quad \theta = \pi - \arcsin\left(\frac{3}{5}\right) \quad \theta = \pi + \arcsin\left(\frac{3}{5}\right)$$

Teacher Note

Q4 note: $\arcsin(3/5)$ gives the Q1 reference angle. Q2 reflection is $\pi - \arcsin(3/5)$. The third option $\pi + \arcsin(3/5)$ lands in Q3 where sine is negative.